Isabella Showman

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Education

B.S., University of Washington, Seattle, WA *2022 – 2026*
Atmospheric Sciences with a climate concentration
Minors in Oceanography, Environmental Studies
GPA 3.92/4.0
Interdisciplinary Honors Program
Dean's List, University of Washington, 12/12 consecutive quarters

Research Experience

Department of Atmospheric and Climate Science, University of Washington
Research Scientist/Engineer Assistant (Advisor: Dr. Alexander Turner) *2026 – present*
- Analyzing Landsat 4-9 and Sentinel-2 satellite data to develop a method that can detect methane permafrost craters in Siberia's Yamal Peninsula
- Developing method to detect changes in surface to indicate crater formation

Undergraduate Research Assistant (Advisor: Dr. Alexander Turner) *2025 – 2026*
- Compared Sentinel-1/2 and Landsat SAR and SWIR remote sensing to investigate abnormal crater features

Applied Ocean Physics & Engineering, Woods Hole Oceanographic Institution
Guest Student Researcher (Advisor: Dr. Tom Bell) *2026*
- Observing sea ice extent off the Northern Alaskan coast through the melting season using Sentinel-1 C-SAR and NASA PACE satellite products
- Detecting how sea ice retreats spatially and in proximity to river mouths

National Aeronautics and Space Administration (NASA), Wallops Flight Facility, VA
NASA Student Airborne Research Program (SARP) Intern (Advisor: Dr. Tom Bell) *2025*
- Developed method for classifying local Alaskan sea ice extent to interpolate freshet event timing
- Interpreted Sentinel-1 C-SAR satellite data to observe sea ice extent
- Flew on NASA's P-3 Orion to collect air samples and chemical measurements over the greater DC area

School of Aquatic and Fishery Sciences, EarthLab, University of Washington
Undergraduate Research Assistant (Advisor: Dr. Tim Essington) *2024 – 2025*
- Analyzed published critical oxygen thresholds of key Puget Sound species from literature in RStudio
- Evaluating threshold ranges and writing a paper summary comparing oxygen thresholds with environmental variables working toward publication

Dissolved Oxygen Vulnerability & Climate Change Intern (Advisor: Dr. Tim Essington) *2024*
- Systematically reviewed ~400 papers to create a database of oxygen thresholds for three key Puget Sound species
- Developed important initial step into a comprehensive vulnerability analysis of the Puget Sound ecosystem to human nutrient inputs

Washington Ocean Acidification Center (WOAC) 5-day Research Cruise, Puget Sound, WA
Research Scientist *2024*
- Collected nutrient, salinity, DIC, and chlorophyll samples from CTD sampler at every station
- Prepared 150+ chlorophyll samples on board immediately after collection for onshore analysis

UK Centre for Ecology and Hydrology
Literature Scholar - Combined Sewage Overflow Paper Intern (Advisor: Andrew Singer) *2023*
- Analyzing 80+ sources online and extracting key quotes to identify UK government response to consistent combined sewer overflows
- Uncovered inconsistencies among government published documents and quotes

Teaching and Leadership Experience

Students Explore Aquatic and Fishery Sciences (SEAS) Board Member *2025 – present*
- Teach lessons to K-12 students related to marine science
- Creating interactive lesson plans to get young students excited about the environment
- Developing lesson related to Arctic sea ice and albedo for high school students

Sustainability Research Lead, University of Washington RSO ReThink *2023 – 2025*
- Designed and developed criteria to advocate for sustainability of local businesses
- Managed smaller research team within club and delegated tasks

Awards

Automated detection and discovery of Siberian methane permafrost eruption craters
- University of Washington Program on Climate Change (PCC) Climate Science Research Acceleration Award
- PI: Alexander Turner; Award: $30,000; 09.2026 – 07.2027

Detecting Coastal Sea Ice Extent and Freshet Event Timing in Prudhoe Bay, Alaska using Sentinel-1 C-SAR
- American Geophysical Union Annual Meeting Travel
  - University of Washington Population Health Initiative; Award: $690; Dec-2025
  - University of Washington College of the Environment Travel & Meeting Fund; Award: $770.79; Dec-2025

Presentations

2026 **Showman, I.,** Turner, A. "Detecting Permafrost Gas Blowout Crater Formation Using Satellite-Derived Water Indices and Changepoint Detection." **Poster** presented at University of Washington Atmospheric and Climate Science Undergraduate Research Symposium, Seattle, WA.

2026 **Showman, I.,** Bell, T., Lang, S., Maksym, T. "Detecting Coastal Sea Ice Extent and Freshet Event Timing in Prudhoe Bay, Alaska using Sentinel-1 C-SAR." **Poster** presented at University of Washington Undergraduate Research Symposium, Seattle, WA.

2025 **Showman, I.,** Bell, T., Lang, S., Maksym, T. "Detecting Coastal Sea Ice Extent and Freshet Event Timing in Prudhoe Bay, Alaska using Sentinel-1 C-SAR." **Poster** presented at the **American Geophysical Union (AGU)**, New Orleans, LA.

2025 **Showman, I.,** Bell, T., Lang, S. "Detecting Coastal Sea Ice Extent and Freshet Event Timing in Prudhoe Bay, Alaska using Sentinel-1 C-SAR." **Oral** presentation at the conclusion of the **NASA SARP** internship, Richmond, VA.

Invited events

2024 Met with a group of Congressional staffers from DC, who directly support my elected federal delegation, to present my work on dissolved oxygen vulnerabilities in the Puget Sound.

2023 "The 8th International Day of Women and Girls in Science Assembly talk on gender equality in sustainable development programs." Presented at UN Headquarters in New York.

2022 One of first university students in North America to participate in Sustain2030 - a German simulation focusing on the UN SDGs.